PYTHON ACTIVITY RECORD

Student Feedback Sentiment Scoring

1. Problem Statement

Create a Python program that reads student feedback, counts predefined positive and negative keywords, calculates a sentiment score, and classifies the feedback as Positive, Negative, or Neutral.

2. Objectives

- Accept feedback text from the user.
- Convert feedback to lowercase and replace selected punctuation with spaces.
- Split feedback into individual words.
- Count matches with predefined positive and negative keyword lists.
- Calculate the sentiment score by subtracting the negative count from the positive count.
- Display the positive count, negative count, score, and sentiment category.

3. Assumptions

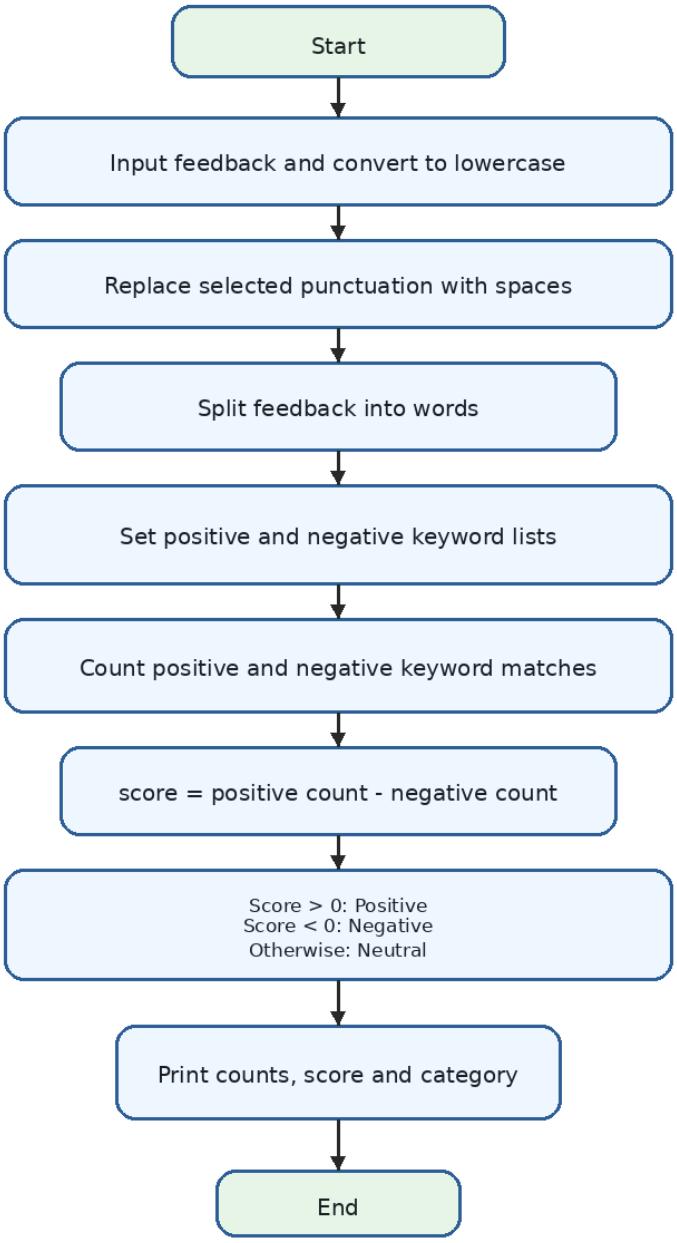
- The user enters feedback as text.
- Only the keywords in the program's lists are counted: positive words are good, helpful, excellent, clear, and useful; negative words are bad, confusing, poor, difficult, and boring.
- Punctuation marks . , ! ? ; : are replaced with spaces.
- Words are matched exactly after converting the feedback to lowercase.
- The score is positive count minus negative count; a score of zero is classified as Neutral.
- This is a simple keyword-based approach and does not understand context, negation, or words not in the lists.

4. Algorithm

1. Start the program.
2. Read the feedback and convert it to lowercase.
3. Replace each selected punctuation mark with a space.
4. Split the feedback into a list of words.
5. Create the positive and negative keyword lists.
6. Count how many words match the positive list and how many match the negative list.
7. Calculate score = positive count - negative count.
8. If score is greater than zero, set category to Positive.
9. If score is less than zero, set category to Negative; otherwise set category to Neutral.
10. Display the positive count, negative count, score, and category.
11. Stop.

5. Flowchart

The flowchart shows how feedback is cleaned, scored, classified, and displayed.



6. Pseudocode

```
BEGIN
    INPUT feedback
    CONVERT feedback to lowercase

    FOR each mark in ".,!?;:"
        REPLACE mark with a space
    END FOR

    words = split feedback into words
    positive = ["good", "helpful", "excellent", "clear", "useful"]
    negative = ["bad", "confusing", "poor", "difficult", "boring"]

    pos = count words that are in positive
    neg = count words that are in negative
```

```
    score = pos - neg

    IF score > 0
        category = "Positive"
    ELSE IF score < 0
        category = "Negative"
    ELSE
        category = "Neutral"
    END IF

    DISPLAY pos, neg, score, category
END
```

7. Python Source Code

```
feedback = input("Feedback: ").lower()

for mark in ".,!?;:":

    feedback = feedback.replace(mark, " ")

words = feedback.split()

positive = ["good", "helpful", "excellent", "clear", "useful"]

negative = ["bad", "confusing", "poor", "difficult", "boring"]

pos = sum(1 for word in words if word in positive)

neg = sum(1 for word in words if word in negative)

score = pos - neg

if score > 0:

    category = "Positive"

elif score < 0:

    category = "Negative"

else:

    category = "Neutral"

print("Positive keywords:", pos)

print("Negative keywords:", neg)

print("Score:", score)

print("Category:", category)
```

8. Test Cases

Test	Feedback input	Expected result	Purpose
1	The class was good, clear, and helpful!	Positive=3, Negative=0, Score=3, Category=Positive	Positive feedback
2	The lecture was bad and confusing.	Positive=0, Negative=2, Score=-2, Category=Negative	Negative feedback

3	The class was okay today.	Positive=0, Negative=0, Score=0, Category=Neutral	Neutral feedback
4	The lesson was GOOD, but boring.	Positive=1, Negative=1, Score=0, Category=Neutral	Mixed feedback

9. Output Screenshots

Test Case 1: Positive sentiment

```
Python Output - Positive sentiment
Feedback: The class was good, clear, and helpful!
Positive keywords: 3
Negative keywords: 0
Score: 3
Category: Positive
```

Test Case 2: Negative sentiment

```
Python Output - Negative sentiment
Feedback: The lecture was bad and confusing.
Positive keywords: 0
Negative keywords: 2
Score: -2
Category: Negative
```

Test Case 3: Neutral sentiment

```
Python Output - Neutral sentiment
Feedback: The class was okay today.
Positive keywords: 0
Negative keywords: 0
Score: 0
Category: Neutral
```

10. Conclusion

The program demonstrates Python string processing, lists, loops, counting, arithmetic operations, and conditional statements. It calculates a simple sentiment score from predefined keywords and labels feedback as Positive, Negative, or Neutral. Because it relies only on exact keyword matches, it may not correctly understand context or negation.